

TESTING AWS ENDPOINTS

Prerequisites

1. AWS Account credentials with Lambda Access [Request Access](#)
2. User account in [Sayaam For All\(test-saayam.netlify.app\)](#)

Testing with AWS Lambda (In-built Testing)

DEMO - [Watch here](#)

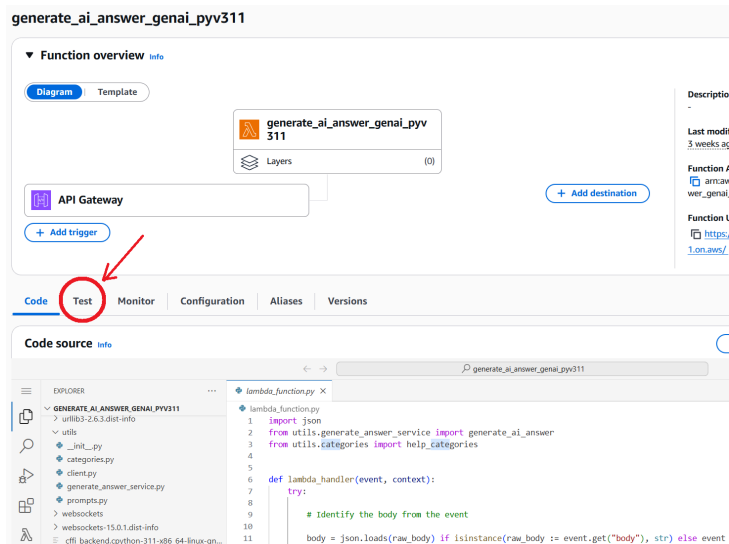
TESTING STEPS

Step 1: Sign In and Navigate to AWS Lambda Function

1. Sign-In into AWS Console using your aws account credentials (aws.amazon.com)
2. Search and navigate to the lambda functions ([Functions | Lambda](#))
3. Lookup the Lambda Function name for testing and open it

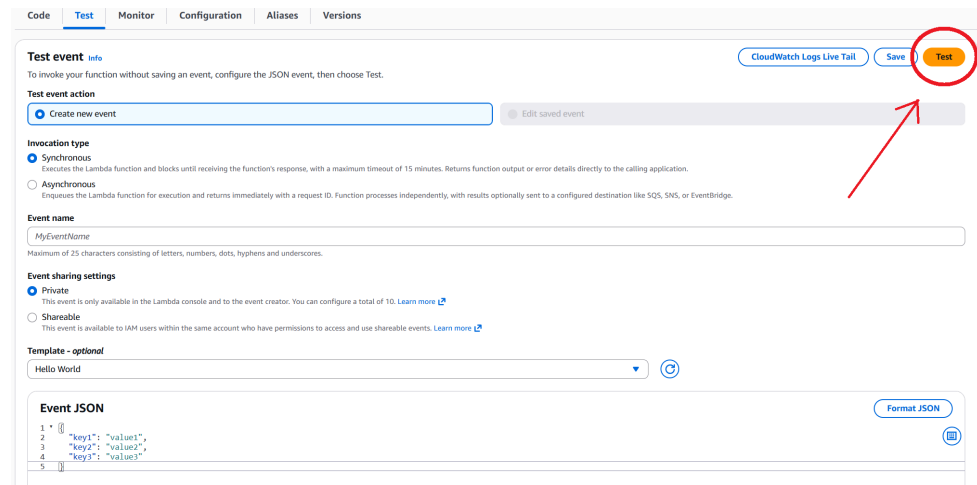
Step 2: Testing the function using the in-built Test Event in AWS Lambda

1. Navigate to the *Test* Tab (Next to *Code*)



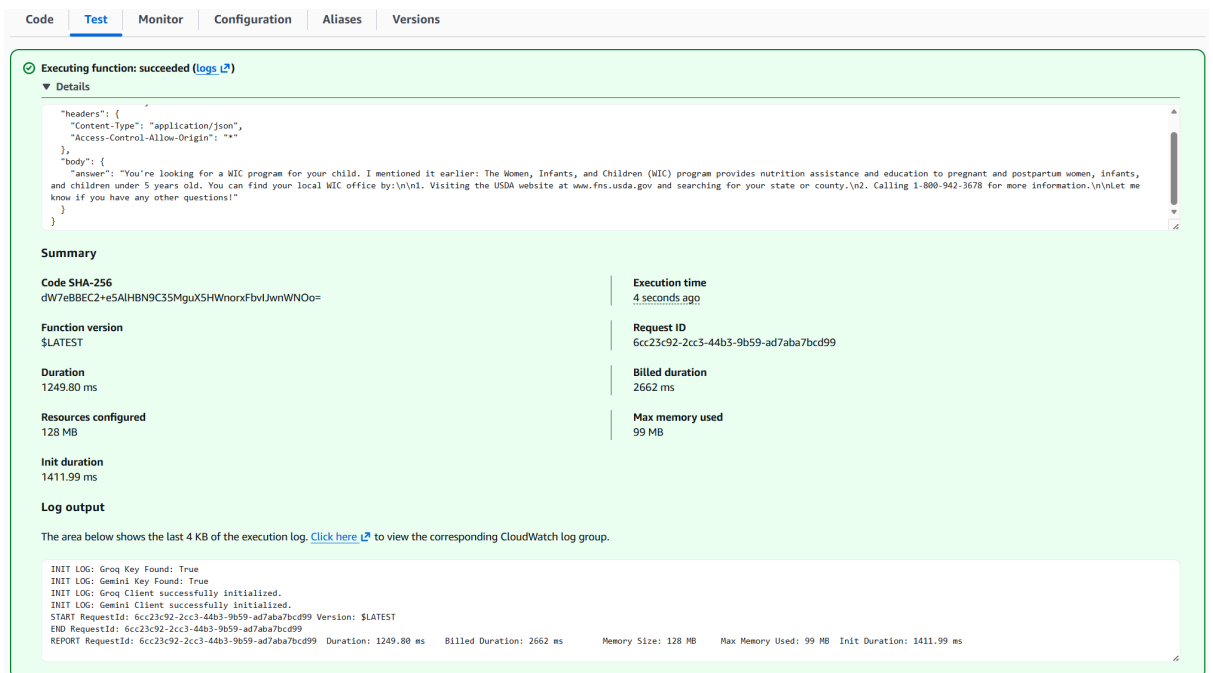
2. Test Event Configuration for Testing API Endpoints on AWS Lambda
 - a. Test Event Action: **Create New Event**
 - b. Invocation Type: **Synchronous**
 - c. Event Name: *(Optional)*
 - d. Event Sharing Settings: **Private**. *Shareable* only when explicitly stated or sharing a particular test case result/anomaly with someone
 - e. Template: *(Optional)*

3. **Event JSON** is the section where you enter the raw request payload that the Lambda function will receive. After adding your JSON body, run the test by selecting **Test** in the upper-right corner of the Test tab.



The screenshot shows the AWS Lambda console's 'Test' tab. At the top, there are tabs for 'Code', 'Test', 'Monitor', 'Configuration', 'Aliases', and 'Versions'. The 'Test' tab is active. In the top right corner, there are buttons for 'CloudWatch Logs Live Tail', 'Save', and 'Test'. The 'Test' button is circled in red, and a red arrow points to it from the right. Below these buttons, there's a section for 'Test event action' with a 'Create new event' button and an 'Edit saved event' button. The 'Invocation type' section has two options: 'Synchronous' (selected) and 'Asynchronous'. The 'Event name' section has a text input field with 'MyEventName'. The 'Event sharing settings' section has two options: 'Private' (selected) and 'Shareable'. The 'Template - optional' section has a dropdown menu with 'Hello World'. The 'Event JSON' section is at the bottom, showing a code editor with a JSON payload. A 'Format JSON' button is next to the code editor.

4. Upon a Successful Execution, An output block similar to this would appear



The screenshot shows the AWS Lambda console's 'Test' tab after a successful execution. The 'Test' button is still visible. Below it, there's a green banner indicating 'Executing function: succeeded (logs 2)'. The 'Details' section is expanded, showing the JSON output of the function. The 'Summary' section shows various execution metrics. The 'Log output' section shows the execution log.

Details

```
{
  "headers": {
    "Content-Type": "application/json",
    "Access-Control-Allow-Origin": "*"
  },
  "body": {
    "answer": "You're looking for a WIC program for your child. I mentioned it earlier: The Women, Infants, and Children (WIC) program provides nutrition assistance and education to pregnant and postpartum women, infants, and children under 5 years old. You can find your local WIC office by:\n\n1. Visiting the USDA website at www.fns.usda.gov and searching for your state or county.\n2. Calling 1-800-942-3678 for more information.\n\nLet me know if you have any other questions!"
  }
}
```

Summary

Code SHA-256 dW7eBBEC2+e5AlHBN9C3SMguX5HWnorrFbv1JwmWNO=	Execution time 4 seconds ago
Function version \$LATEST	Request ID 6cc23c92-2cc3-44b3-9b59-ad7aba7bcd99
Duration 1249.80 ms	Billed duration 2662 ms
Resources configured 128 MB	Max memory used 99 MB
Init duration 1411.99 ms	

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
INIT LOG: Groq Key Found: True
INIT LOG: Gemini Key Found: True
INIT LOG: Groq Client successfully initialized.
INIT LOG: Gemini Client successfully initialized.
START RequestId: 6cc23c92-2cc3-44b3-9b59-ad7aba7bcd99 Version: $LATEST
END RequestId: 6cc23c92-2cc3-44b3-9b59-ad7aba7bcd99
REPORT RequestId: 6cc23c92-2cc3-44b3-9b59-ad7aba7bcd99 Duration: 1249.80 ms Billed Duration: 2662 ms Memory Size: 128 MB Max Memory Used: 99 MB Init Duration: 1411.99 ms
```

Testing Cognito Authorized API Gateway Endpoints (Using Postman)

DEMO - [Watch here](#)

These API endpoints are:

- Hosted on **AWS Lambda**
- Exposed via **AWS API Gateway**
- Secured using a **Cognito User Pool Authorizer**
- Require a valid **Cognito ID Token (JWT)** for access

Note on Authentication

When a user logs in through the application, **Amazon Cognito** authenticates the user and issues a **JSON Web Token (JWT)**.

This JWT:

- Represents the authenticated user
- Contains user identity claims (such as user ID and email)
- Is required to access protected API endpoints

The API endpoints are secured using a Cognito Authorizer in **Amazon API Gateway**, which validates this JWT before allowing access. Therefore, to test the APIs, we must retrieve the issued **ID Token (JWT)** and include it in the request as a Bearer Token.

TESTING STEPS

RETRIEVING ID TOKEN (JWT)

Step 1: Log in to Saayam Website

1. Open site : <https://test-saayam.netlify.app/>
2. Create your account
3. Log in with valid user credentials.

**In this step - We are logging into the test site, based on your business case, login to the site you wish.*

Step 2: Open Developer Tools

1. Right-click anywhere on the page.
2. Click **Inspect**
3. Go to the **Application** tab (Chrome/Edge).

Step 3: Locate the ID Token

-
- The screenshot shows a web browser's developer tools interface. The 'Local storage' tab is active, displaying a list of key-value pairs. The key 'CognitoidentityServiceProvider.58tba3bjsk2v90kavksbqptrm.24788438-e021-7092-216d-39f46e671c29.idToken' is selected, and its value, 'eyJraWQ1OjI2SGxhT1VUaW5kdTkr cmZwN1VwaGVXGTJrImcl203MkZlMwK2==', is shown in the right-hand pane. A blue arrow points to the 'idToken' entry in the list.
- | Key | Value |
|--|--|
| CognitoidentityServiceProvider.58tba3bjsk2v90kavksbqptrm.24788438-e021-7092-216d-39f46e671c29.deviceG... | a6fcf711-b6b5-40a2-9f0f-19839245ceb2 |
| CognitoidentityServiceProvider.58tba3bjsk2v90kavksbqptrm.24788438-e021-7092-216d-39f46e671c29.deviceK... | us-east-1-ad6dc292-5f54-477b-bbef-47053c8864ad |
| CognitoidentityServiceProvider.58tba3bjsk2v90kavksbqptrm.24788438-e021-7092-216d-39f46e671c29.idToken | eyJraWQ1OjI2SGxhT1VUaW5kdTkr cmZwN1VwaGVXGTJrImcl203MkZlMwK2== |
| CognitoidentityServiceProvider.58tba3bjsk2v90kavksbqptrm.24788438-e021-7092-216d-39f46e671c29.random... | 0wPjP5/Kegg9PvIdAbXThR9XKCqDbHMP8mfDAC20zHoryqWXAiKkA== |

Common Issues & Troubleshooting

1. Token Expired

Cognito ID tokens typically expire after 1 hour.

Solution: Log in again and retrieve a fresh token.

2. Using Wrong Token Type

Make sure you are using:

-  ID Token
-  Not Access Token
-  Not Refresh Token